Title of the Project

FACIAL RECOGNITION USING PCA (EIGENFACES)

Capstone-I project report submitted

by the student of

Hybrid UG program in Artificial Intelligence & Cyber Security

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Declaration

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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and Signature: Omprakash Kumar.

Content

Summary of the Project

This project focuses on facial recognition using Principal Component Analysis (PCA), specifically the Eigenfaces method. The aim is to reduce the dimensionality of facial images while retaining important features for identification. The system involves preprocessing grayscale images, computing the mean face, and extracting principal components from the dataset. Each face is then projected onto a lower-dimensional eigenspace for efficient representation. Recognition is performed by comparing feature vectors using distance measures. The project uses the AT&T ORL (Olivetti) face dataset consisting of 400 images of 40 individuals. The current work emphasizes understanding and implementing PCA from scratch for facial recognition.

Mention the work done by each member:

1.Omprakash Kumar: Responsible for mathematical formulation and implementation of the core PCA (Eigenfaces) algorithm, including mean face computation, covariance matrix formulation, eigenvalue decomposition, and dimensionality reduction.

2.Nitish Kumar: Responsible for image preprocessing, dataset handling, and implementation of the recognition pipeline, including projection of images into eigenspace and similarity measurement using distance metrics.

3.Nilesh Kumar: Responsible for visualization, performance evaluation, and system integration, including plotting

eigenfaces, analysing accuracy, and preparing presentation and documentation.

1. **Introduction**:

Facial recognition is an important application of computer vision used in security and authentication systems. This project aims to develop a face recognition system using Principal Component Analysis (PCA) through the Eigenfaces method. PCA reduces the dimensionality of facial images while preserving important features for identification. The system is implemented using the AT&T ORL face dataset. The main focus of this work is to understand the mathematical concepts of PCA and apply them to facial recognition.

2. Literature review:

The Eigenfaces method for face recognition was introduced by Matthew Turk and Alex Pentland in 1991. It is based on Principal Component Analysis (PCA), a statistical technique used for dimensionality reduction. PCA identifies directions of maximum variance in the data and represents images in a lower-dimensional space. This method has been widely used for face recognition due to its simplicity and efficiency. However, it has limitations under varying lighting conditions and facial expressions. Despite these limitations, Eigenfaces remain a fundamental approach for understanding face recognition systems.

3. **Methodology:**

- Data preprocessing
- PCA steps
- Recognition method

4. **Implementation:**

- Python
- NumPy
- OpenCV
- Steps of code

5. **Result:**

- Accuracy
- Eigenfaces visualization
- Graphs

6. **Conclusion:**

This project demonstrates the implementation of facial recognition using Principal Component Analysis (PCA) and the Eigenfaces method. The approach effectively reduces the dimensionality of facial images while preserving significant features required for recognition. The system highlights how mathematical concepts such as eigenvalues and eigenvectors can be applied in real-world computer vision problems. The results show that PCA can be used as a simple and efficient technique for face recognition under controlled conditions.

7. Future work:

Future work can include improving the robustness of the system under varying lighting and pose conditions. Advanced techniques such as Linear Discriminant Analysis (LDA) and Convolutional Neural Networks (CNNs) can be explored for better performance. Real-time face recognition using webcam input and integration with security systems can also be implemented. Additionally, addressing challenges such as illumination variation and dataset bias can further enhance the system.

Chapter 1

Introduction:

Facial recognition is an important application of computer vision used in security, authentication, and surveillance systems. This project aims to develop a face recognition system using Principal Component Analysis (PCA), specifically the Eigenfaces method. PCA reduces the dimensionality of facial images while preserving important features required for identification. In this approach, facial images are transformed into a lower-dimensional space where recognition is performed efficiently. The system is implemented using the AT&T ORL face dataset. The primary objective of this work is to understand the mathematical foundation of PCA and apply it to facial recognition.

Chapter 2:

The Eigenfaces method for face recognition was introduced by Matthew Turk and Alex Pentland in 1991. It is based on Principal Component Analysis (PCA), a statistical technique used for dimensionality reduction. PCA identifies directions of maximum variance in the data and represents images in a lower-dimensional space. This method has been widely used for face recognition due to its simplicity and efficiency. However, it has limitations under varying lighting conditions and facial expressions. Despite these limitations, Eigenfaces remain a fundamental approach for understanding face recognition systems.

Chapter 3:

Methodology:

3.1 Image representation:

Each facial image is converted into a grayscale image and resized to a fixed dimension of 92 × 112 pixels. The image is then flattened into a one-dimensional vector of size 10304. All images are combined to form a data matrix where each row represents one image and each column represents a pixel value.

3.2 Mean Face Computation:

The mean face is computed by averaging all training images. It represents the average appearance of all faces in the dataset. This mean face is later used to center the data by subtracting it from each image.

Chapter 3:

Mean Face

Mean Subtracted Face

Chapter 3:

3.3 Mean Subtraction:

Each image is normalized by subtracting the mean face. This step ensures that PCA captures variations in facial features rather than absolute pixel intensity values. The resulting images represent deviations from the average face.

3.4 Covariance Matrix and Dimensionality Reduction:

The covariance matrix is computed to analyze the variance structure of the dataset. Instead of computing a large covariance matrix in pixel space, a smaller matrix is calculated using the relation $A A^T$, where A is the mean-centered data matrix. This reduces computational complexity significantly.

Chapter 3:

3.5 Eigenfaces Computation:

Eigenvalues and eigenvectors of the covariance matrix are computed. The eigenvectors are then mapped back to the original image space to obtain eigenfaces. These eigenfaces represent the principal components of the dataset and capture the most significant variations among faces.

3.6 Projection and Recognition:

Each face image is projected onto a lower-dimensional eigenspace using the selected eigenfaces. Recognition is performed by comparing the projected test image with projected training images using Euclidean distance. The closest match is considered the recognized face.

Chapter 4:

Implementation:

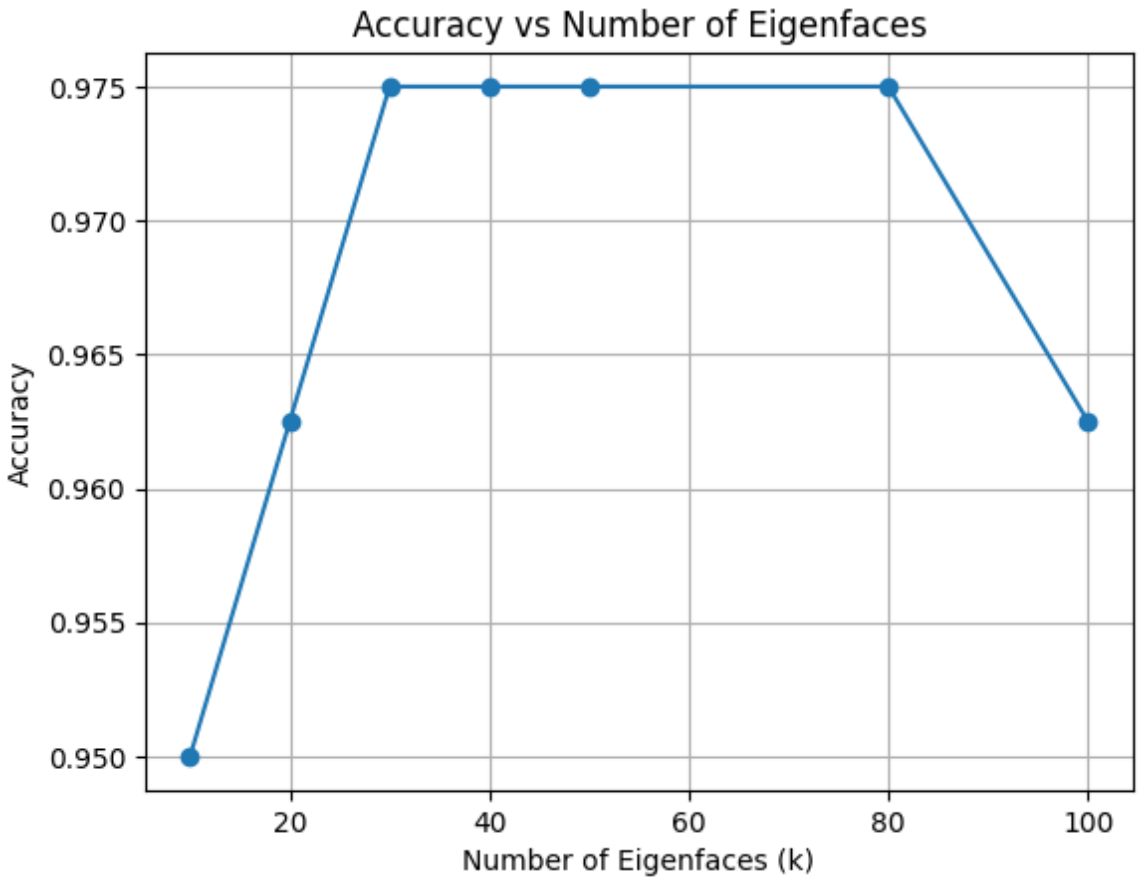
The system is implemented using Python with libraries such as NumPy, OpenCV, and Matplotlib. The AT&T ORL dataset is used, containing 400 grayscale images of 40 individuals. The dataset is divided into training and testing sets, with 8 images per subject used for training and 2 for testing. PCA is implemented from scratch, including mean computation, covariance calculation, eigen decomposition, and projection. The system performs recognition using a distance-based classification approach.

Chapter 5:

Result and Analysis:

The performance of the system is evaluated using a train-test split of the dataset. The model achieved an accuracy of approximately 97.5% using 50 principal components. It is observed that the recognition accuracy depends on the number of eigenfaces used. Smaller values of k result in loss of important information, while larger values may introduce noise. An optimal value of k provides the best balance between accuracy and dimensionality reduction. The eigenfaces generated appear as ghost-like images, representing dominant patterns such as lighting variations and facial structure.

- Accuracy graph.



It is observed that the accuracy increases as the number of eigenfaces (k) increases initially, as more relevant facial features are captured. However, beyond a certain point, the accuracy stabilizes and may slightly decrease due to inclusion of noise and less significant components. This demonstrates the importance of selecting an optimal value of k for best performance.

Chapter 6:

Conclusion:

This project successfully demonstrates facial recognition using Principal Component Analysis and the Eigenfaces method. The system effectively reduces dimensionality while preserving important facial features. The results show that PCA can achieve high recognition accuracy under controlled conditions. The project also highlights the importance of mathematical concepts such as eigenvalues and eigenvectors in real-world applications.

Chapter 7:

Future Work:

Future improvements can include handling variations in lighting, pose, and facial expressions. Advanced techniques such as Linear Discriminant Analysis (LDA) and deep learning-based methods like Convolutional Neural Networks (CNNs) can be explored for improved performance. Real-time face recognition using webcam input and enhanced security features can also be implemented.

References

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